any trouble we would put in the right amounts of c so that the units would straighten out in the last equation, but not in the intermediate ones.

Thus our equations for energy and momentum are

$$\begin{aligned} E &= m = m_0/\sqrt{1-v^2}, \\ \mathbf{p} &= m\mathbf{v} = m_0\mathbf{v}/\sqrt{1-v^2}. \end{aligned} \tag{5.6}$$

Also in these units, we have

$$E^2 - p^2 = m_0^2. \tag{5.7}$$

For example, if we measure energy in electron volts, what does a mass of 1 electron volt mean? It means the mass whose rest energy is 1 electron volt, that is, m_0c^2 is one electron volt. For example, the rest mass of an electron is 0.511×10^6 ev.

Now what would the momentum and energy look like in a new coordinate system? To find out, we shall have to transform Eq. (5.6), which we can do because we know how the velocity transforms. Suppose that, as we measure it, an object has a velocity v, but we look upon the same object from the point of view of a spaceship which itself is moving with a velocity u, and in that system we use a prime to designate the corresponding thing. In order to simplify things at first, we shall take the case that the velocity v is in the direction of u. (Later, we can do the more general case.) What is v', the velocity as seen from the spaceship? It is the composite velocity, the "difference" between v and u. By the law which we worked out before,

$$v' = \frac{v-u}{1-uv}. \tag{5.8}$$

Now let us calculate the new energy E', the energy as the fellow in the spaceship would see it. He would use the same rest mass, of course, but he would use v' for the velocity. What we have to do is square v', subtract it from one, take the square root, and take the reciprocal: